

Financial Modelling for Gambling Games

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December 20, 2025

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Chapter 1

Introduction

In this document I further my analysis of the games "Flip it" and "Crack It". In "Flip it" I rerun the models with 1 million rounds played, compared to the 10,000 rounds in previous modeling. I also model the effects of a 100x Jackpot to test its viability. In "Crack It" I test to make sure that a whole number of scratchcards are offered and I work out how to decide exactly how many cards are given in each prize size.

Chapter 2

Flip It

2.1 Making Profit

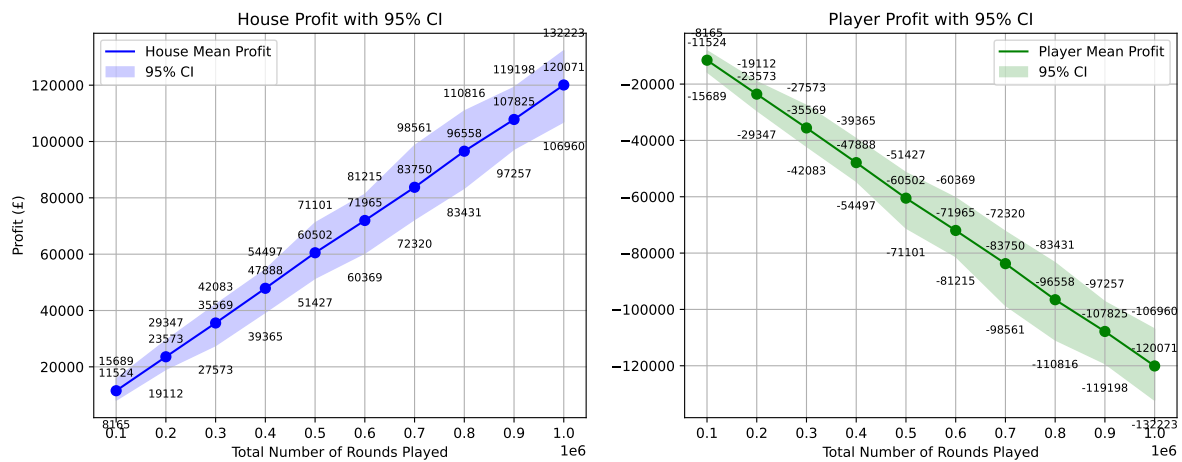


Figure 2.1: House/Player Odds: 51%/49%

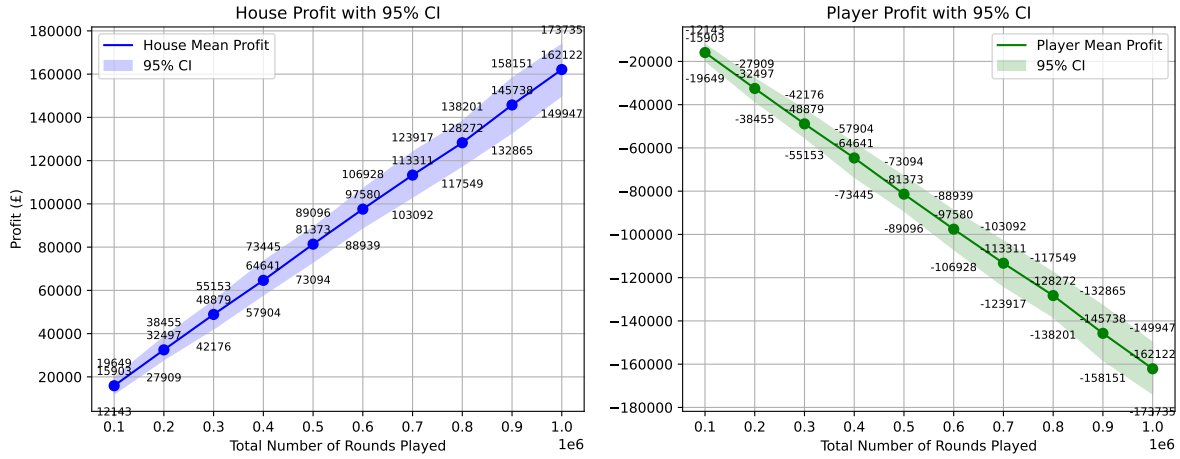


Figure 2.2: European Roulette House/Player Odds: 51.35%/48.65%

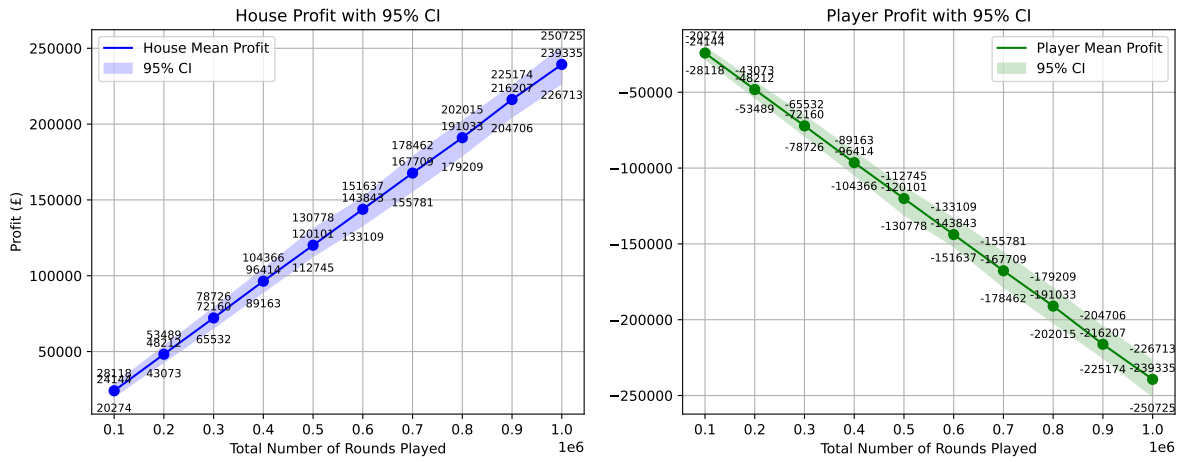


Figure 2.3: House/Player Odds: 52%/48%

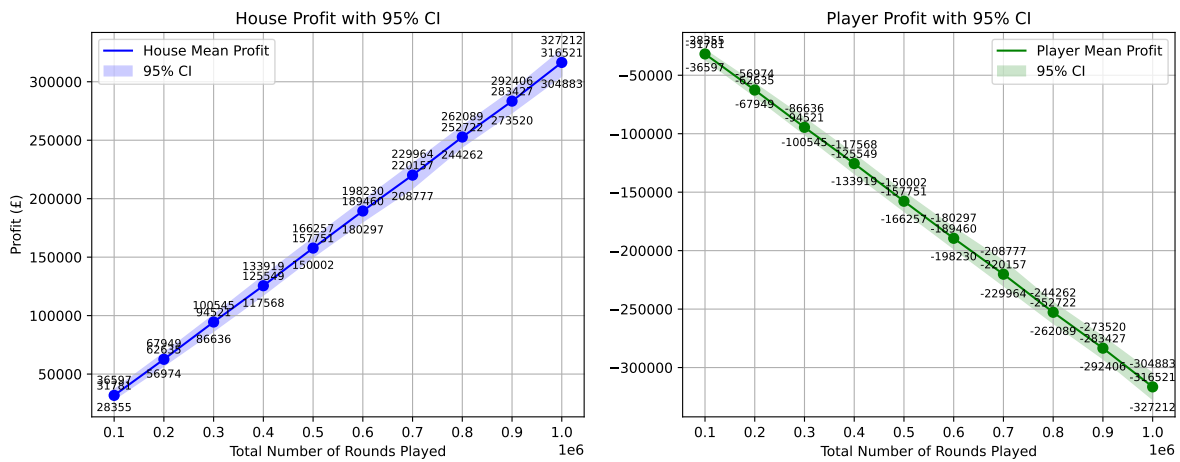


Figure 2.4: European Roulette House/Player Odds: 52.63%/47.37%

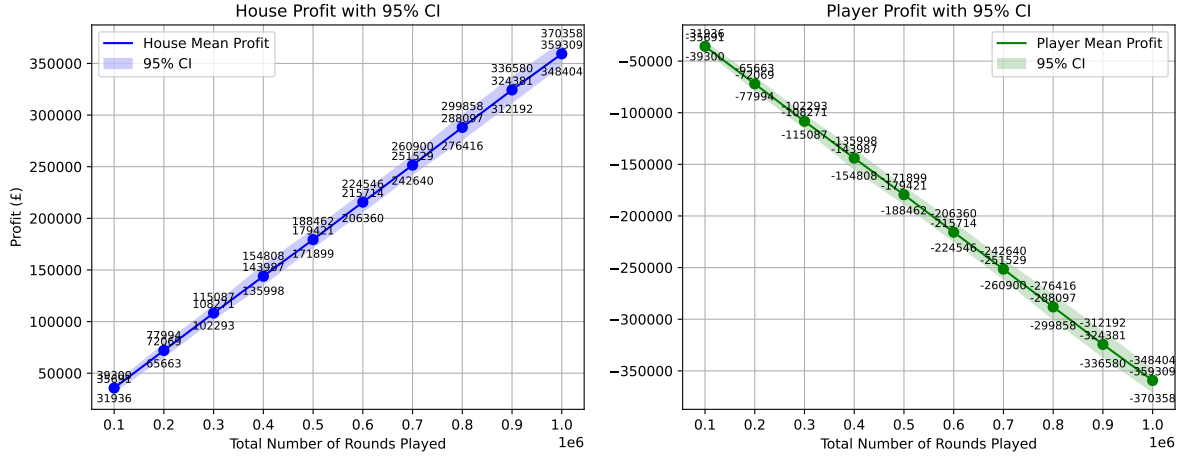


Figure 2.5: House/Player Odds: 53%/47%

The expected return in addition to the 95% confidence intervals for each set of odds is given in Table 2.1 and Figure 2.10 below:

Odds (House:Player)	Expected Return	95% CI (Lower Bound)	95% CI (Upper Bound)
51:49	£120,071	£106,960	£132,223
51.35:48.65	£162,122	£149,947	£173,735
52:48	£239,335	£226,713	£250,725
52.63:47.37	£316,521	£304,883	£327,212
53:47	£359,309	£348,404	£370,358

Table 2.1: Expected Return for different House Odds after 1 million rounds played

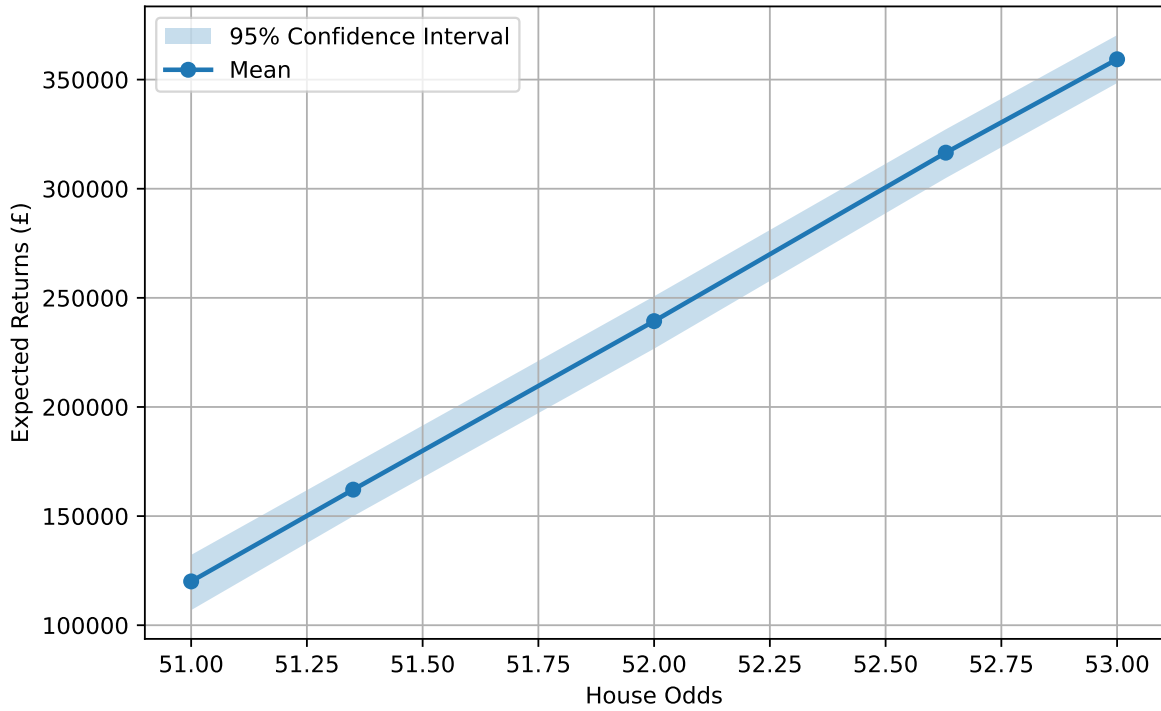


Figure 2.6: Expected Return for different House Odds after 1 million rounds played

This demonstrates a linear increase in expected profit and confidence intervals as the odds change in favour of the house. At 51%, the house is expected to make, on average, Between £0.106 and £0.132 per coin flip. The at 51.35% (European Roulette Odds) this increases to between £0.150 and £0.174 per coin flip, then between £0.227 and £0.251 at 52%, between £0.305 and £0.327 at 52.63% (American Roulette Odds), and between £0.348 and £0.370 at 53%.

2.2 Adding a 100x Jackpot

The following graphs show the expected returns for both the house and player when introducing a 100x Jackpot. It also shows the 95% Lower Bound of returns for the house and 95% Upper Bound of returns for the player. The simulation runs for a total of 100,000 rounds.

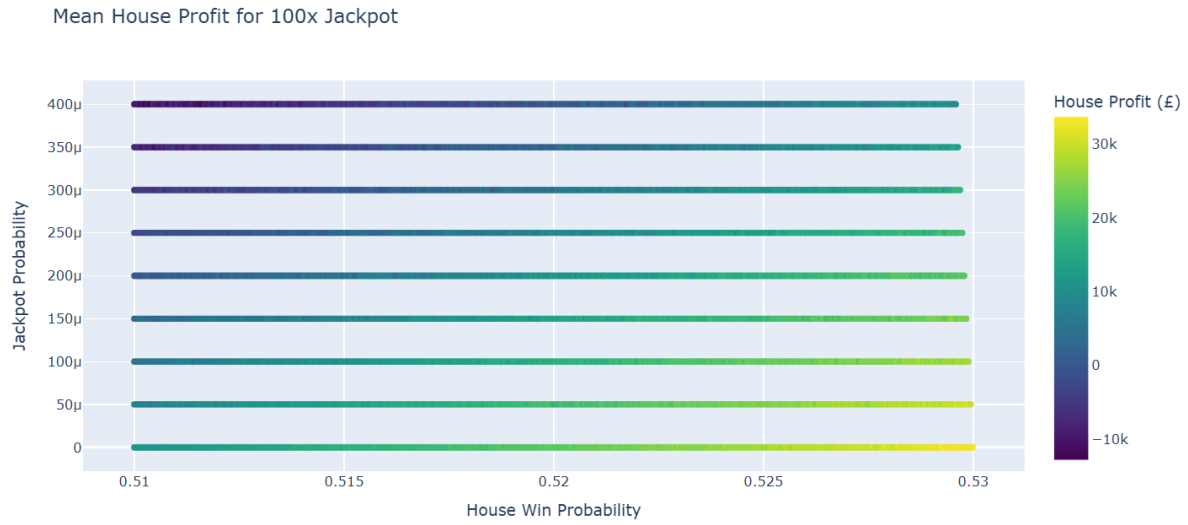


Figure 2.7: Mean House Return for different Jackpot Odds after 100,000 rounds played

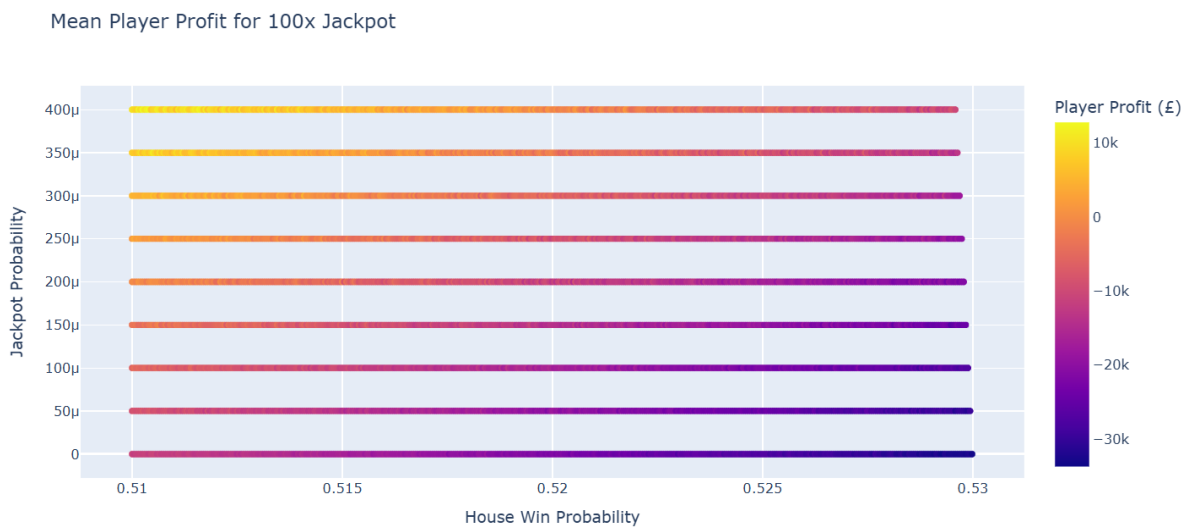


Figure 2.8: Mean Player Return for different Jackpot Odds after 100,000 rounds played

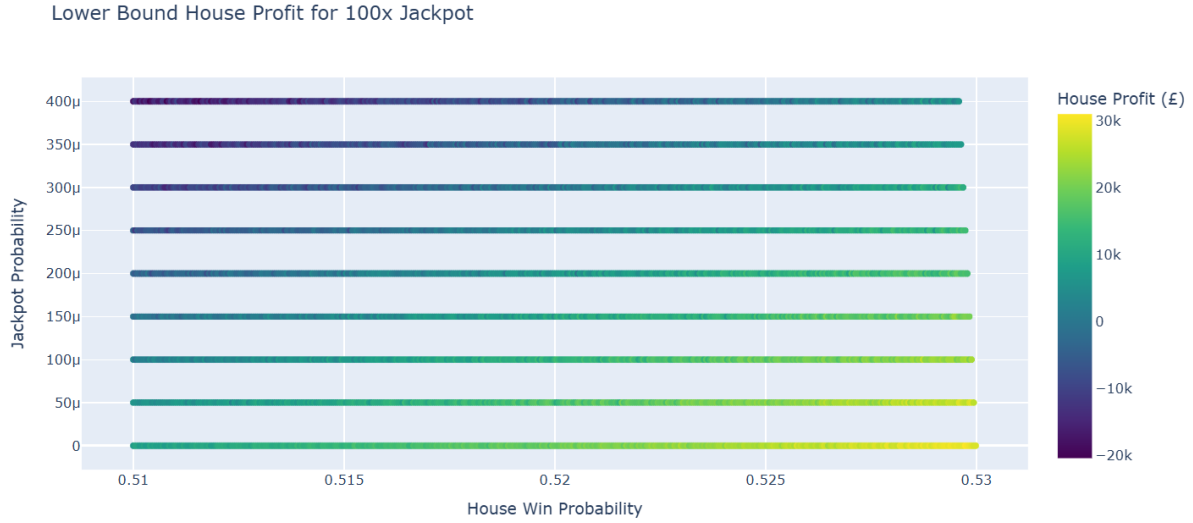


Figure 2.9: Lower Bound House Return for different Jackpot Odds after 100,000 rounds played

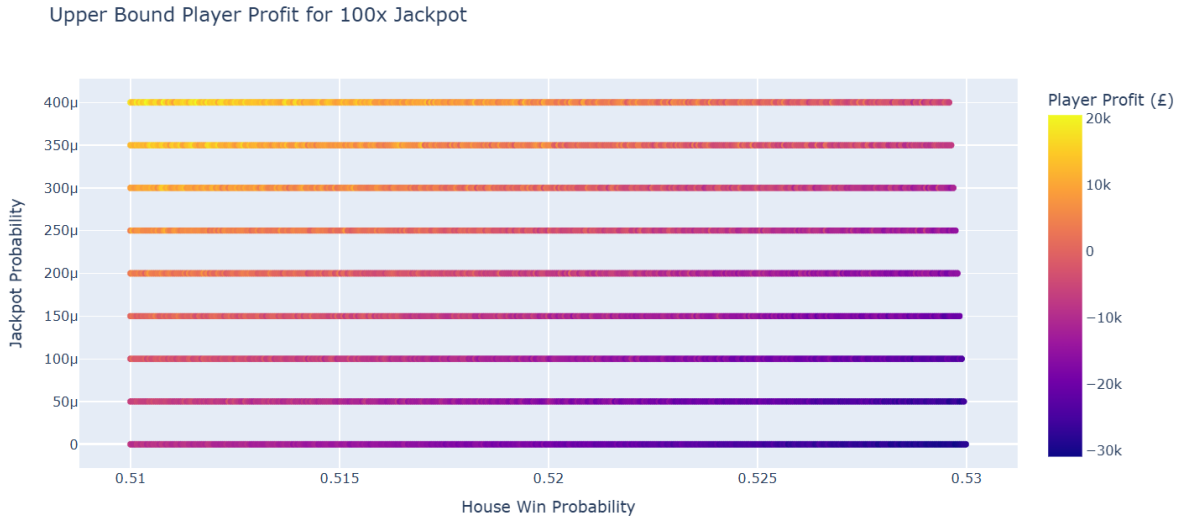


Figure 2.10: Upper Bound Player Return for different House Odds after 1 million rounds played

It is important that we are profiting from this. especially because this model doesn't account for any external costs related to the setting up and running of the game.

The graphs clearly demonstrate the significant effect of adding a 100x jackpot on the house profits. Any jackpot of that scale would need probability of occurrence of 0.0004 at most. With a probability of 0.0004, the house odds would need to be set at at least 52.75% to minimise the risk of losing money, as the simulation demonstrates that the lower bound of house profit continues to give occasional negative values up to that probability. In practice, I believe it would be smarter to set the probability of a jackpot at around 0.0002 and have the houses probability set at around 52% to give good profit and protect against the potential of

net loss. At this setting the mean profit from the simulation was £13435.747 and the lower bound of the 95% Confidence Interval was £7955.069.

Chapter 3

Crack It

Here we update the approach taken in the original document so that it is more generalised:

- There is opportunity to choose how many scratchcard groups there are and how many cards to give each group.
- There is opportunity to choose how many different Prize Levels there are and what the value of each prize level should be.

3.1 Set Up

We denote the following:

Entry fee (y): The fee paid by each player to take part in the game

Game participation (g): The number of players

Payout rate (x): The percentage of the total pot that is distributed between the players

Platform margin (h) = 1 - Payout rate = $1 - x$

Charitable rate (c) = $\frac{1}{10} * \text{Platform margin} = \frac{1}{10} * h$

Then we have:

Total pot = Game participation * Entry fee = $g * y$

Player pot $\approx$ Total pot * Payout Rate = $g * y * x$

Charity pot $\approx$ Total pot * Charitable Rate = $g * y * c$

House Profit $\approx$ (Total pot * Platform margin) - Charity pot = $(g * y * h) - (g * y * c) = (g * y) * (0.9 * h)$

We use approximation symbols as these values are altered slightly to ensure the smooth distribution of prizes to players.

3.2 Scratchcards

The House chooses how many groups to split the participants into and the number of scratchcards to give a player in each group respectively.

We denote the groups $\gamma = [\gamma_1, \dots, \gamma_N]$ where $\sum_{i=1}^N \gamma_i = 1$. In other words the value of each γ_i reflects the size of the percentile group it encapsulates. Then the number of scratchcards to give each player in each group is denoted by $\alpha = [\alpha_1, \dots, \alpha_N]$.

From this we can calculate the total number of scratchcards given to any given group by the following:

$$\text{Total number of Scratchcards} = \sum_{i=1}^N G_i$$

where $G_i = \text{round}(\text{group size} \times \text{game participants} \times \text{scratchcards per player in group}) = \text{round}(\gamma_i * g * \alpha_i)$

In the example below we have $\gamma = [5, 15, 30, 50]$ and $\alpha = [10, 6, 3, 1]$. In other words there are four scratchcard groups, with participants in these groups given 10, 6, 3, and 1 scratchcards respectively.

Percentile	Scratchcards per player
0 – 5%	10
5 – 20%	6
20 – 50%	3
50 – 100%	1

From this we can calculate the total number of scratchcards required.

```
# number of scratchcards

# G1 = 0 - 5 %
# G2 = 5 - 20 %
# G3 = 20 - 50 %
# G4 = 50 - 100 %

G1 = round(Fraction(5, 100) * game_participation) * 10 #10 cards per person
G2 = round(Fraction(15, 100) * game_participation) * 6 #6 cards per person
G3 = round(Fraction(30, 100) * game_participation) * 3 #3 cards per person
G4 = round(Fraction(50, 100) * game_participation) #everyone gets 1

total_scratchcards = G1 + G2 + G3 + G4
```

Figure 3.1: Calculating Total Number of Scratchcards

3.3 Prizes

Prizes are modeled to be multiples of the entry fee in order to ensure consistency with changing entry fees and to aid with the smooth distribution of the Player Pot to the players.

We denote the prize values as $P = [P_1, \dots, P_M]$. The values are determined by $P_i = \beta_i * y$, where $\beta = [\beta_1, \dots, \beta_M]$ are the multiplying coefficients of the entry fee, y . Note that these coefficients should be integer valued in order for the model to work smoothly.

In the example below we have $P = [P_1, P_2, P_3, P_4, P_5]$ and $\alpha = [1, 2, 4, 10, 20]$. In other words there are five categories of prizes with the multiplication of each prize level being 1,2,4,10 and 20 respectively.

Prize Level	Prize Value
Prize 1 (P_1)	y
Prize 2 (P_2)	$2 * y$
Prize 3 (P_3)	$4 * y$
Prize 4 (P_4)	$10 * y$
Prize 5 (P_5)	$20 * y$

3.4 Odds

Having defined the prize values, we need to then calculate the number of each prize to give out in order to fairly and fully distribute the prize pot between the players.

We now define the following:

$$\begin{aligned}
\textbf{Player pot} &= \text{round}(\text{Game participation} * \text{Payout rate}) * \text{Entry fee} \\
&\approx \text{Total pot} * \text{Payout Rate} \\
&= g * y * x
\end{aligned}$$

Scratchcard win % (z): The percentage chance of a scratchcard being a winner

Total winning scratchcard (Z) = round($z * \text{Total scratchcards}$)

Entry fee multiplier (Q) = round($\text{Game participation} * \text{payout rate}$) = round($g * x$)

With these values we now want to make sure that the total number of prizes we give out is equal to **Z** and the sum of each prize value multiplied by the number of them given out is equal to the **Player pot**.

We denote the number of prizes awarded as $N = [N_1, \dots, N_N]$ where N_i is the number of prizes at prize level i . We can then write the first condition as:

$$\sum_{i=1}^N N_i = Z \quad (3.1)$$

The second condition can be written as:

$$\begin{aligned}
&\sum_{i=1}^N P_i * N_i = \text{Player pot} \\
\Rightarrow \sum_{i=1}^N \beta_i * y * N_i &= \text{round}(g * x) y \\
\Rightarrow \sum_{i=1}^N \beta_i * N_i &= Q
\end{aligned} \quad (3.2)$$

We use our example from previously. The app has 72,315 users and "Crack It" is being played by 20% of these players. The entry fee is set at £3.25 and the Payout rate is set at 68%. Then the first condition would be written as:

$$N_1 + N_2 + N_3 + N_4 + N_5 = 4049 \quad (3.3)$$

The second condition would be written as:

$$\begin{aligned} P_1 N_1 + P_2 N_2 + P_3 N_3 + P_4 N_4 + P_5 N_5 &= \text{Player pot} \\ \Rightarrow y N_1 + 2y N_2 + 4y N_3 + 10y N_4 + 20y N_5 &= \text{round}(g \cdot x) y \\ \Rightarrow N_1 + 2N_2 + 4N_3 + 10N_4 + 20N_5 &= 9835 \end{aligned} \quad (3.4)$$

Solving these simultaneous equations will ensure that prizes are distributed in such a way that each flash card has a chance of winning approximately equal to that which has been chosen by The House and that the Player pot, which is a fraction of the Total pot approximately equal to the Payout rate, will be fully distributed to The Players.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import random
import statistics
import math
from fractions import Fraction

# Set the seed
np.random.seed(42)
random.seed(42)sn
```

```
# Set the variables
players = 72315
participation_rate = Fraction(1, 5)
game_participation = players * participation_rate
entry_fee = Fraction(13, 4) # 3.25
payout_rate = Fraction(68, 100)
platform_margin = 1 - payout_rate
charitable_rate = Fraction(1, 10)

print(f"Game Participation: {game_participation}")
```

Game Participation: 14463

```
# Pot values
total_pot = game_participation * entry_fee

player_pot = round(game_participation * payout_rate) * entry_fee
leftover_pot = total_pot - player_pot
charity_pot = leftover_pot * charitable_rate
platform_profit = leftover_pot - charity_pot

print(f"Total pot: £{total_pot:.2f}")
print(f"Player pot: £{player_pot:.2f}")
print(f"Charity pot: £{charity_pot:.2f}")
print(f"Platform profit: £{platform_profit:.2f}")
```

Total pot: £47004.75
 Player pot: £31963.75
 Charity pot: £1504.10
 Platform profit: £13536.90

```

# number of scratchcards

# G1 = 0 - 5 %
# G2 = 5 - 20 %
# G3 = 20 - 50 %
# G4 = 50 - 100 %

G1 = round(Fraction(5, 100) * game_participation) * 10 #10 cards per person
G2 = round(Fraction(15, 100) * game_participation) * 6 #6 cards per person
G3 = round(Fraction(30, 100) * game_participation) * 3 #3 cards per person
G4 = round(Fraction(50, 100) * game_participation) #everyone gets 1

total_scratchcards = G1 + G2 + G3 + G4

print(f"G1: {G1}")
print(f"G2: {G2}")
print(f"G3: {G3}")
print(f"G4: {G4}")
print(f"Total Scratchcards: {total_scratchcards}")

```

G1: 7230
 G2: 13014
 G3: 13017
 G4: 7232
 Total Scratchcards: 40493

```

# Prize Values

P1 = entry_fee * 1
P2 = entry_fee * 2
P3 = entry_fee * 4
P4 = entry_fee * 10
P5 = entry_fee * 20

print(f"P1: £{P1:.2f}")
print(f"P2: £{P2:.2f}")
print(f"P3: £{P3:.2f}")
print(f"P4: £{P4:.2f}")
print(f"P5: £{P5:.2f}")

```

P1: £3.25
 P2: £6.50
 P3: £13.00
 P4: £32.50
 P5: £65.00

```

z = Fraction(1, 10)
total_winning_scratchcards = round(z * total_scratchcards)
Val=round(game_participation * payout_rate)

print(f"Total Winning Scratchcards: {total_winning_scratchcards}")
print(f"Entry Fee Multiplier: {Val}")

```

Total Winning Scratchcards: 4049
 Entry Fee Multiplier: 9835